2025.12.1 (Chain Rule & Sobolev Space)

Thm1 If $f\in C^1(\mathbb{R})$, $f'\in L^\infty(\mathbb{R})$, $u(x)\in W^1(\Omega)$, then

$f(u(x))\in W^1(\Omega)$ & weak $\partial_{x_i}(f(u(x))) = f'(u(x))\,\partial_{x_i}u$, $i=1\cdots n$.

i.e. weak $\nabla f(u) = f'(u)\nabla u$.

Pf: Since $u\in W^1(\Omega)$, by Thm1, $\exists\{u_k\}_{k=1}^\infty\subset C^\infty(\Omega)$ s.t.

$u_k\to u$ in $L^1_{loc}(\Omega)$ $k\to\infty$

$\nabla u_k\to\nabla u$ in $(L^1_{loc}(\Omega))^n$

Consider $\{f(u_k)\}_{k=1}^\infty\subset C^1(\Omega)$:

$\{f(u_k)\}_{k=1}^\infty\subset L^1_{loc}(\Omega)$ & $f(u)\in L^1_{loc}(\Omega)$

Since $\forall\Omega'\subset\subset\Omega$, $\int_{\Omega'}|f(u)|dx\le\int_{\Omega'}|f(u)-f(0)|+|f(0)|dx$

$=\int_{\Omega'}|f'(\xi)||u|+|f(0)|dx<\infty$

$\int_{\Omega'}|f(u_k)-f(u)|dx=\int_{\Omega'}|f'(\xi)||u_k-u|dx\le\|f\|_{L^\infty}\int_{\Omega'}|u_k-u|dx\to0$

$\int_{\Omega'}|f'(u_k)\nabla u_k-f'(u)\nabla u|dx\le\int_{\Omega'}|f'(u_k)\nabla u_k-f'(u_k)\nabla u|dx+\int_{\Omega'}|f'(u_k)\nabla u-f'(u)\nabla u|$

$\le\|f\|_{L^\infty}\int_{\Omega'}|\nabla u_k-\nabla u|dx\to0$ $(k\to\infty)$ By DCT (BCT)

Since $u_k\to u$ in $L^1_{loc}(\Omega)$, $\exists k_j$ s.t. $u_{k_j}\to u$ a.e. $(j\to\infty)$.

$|f'(u_k)-f'(u)||\nabla u|\le2\|f\|_{L^\infty}|\nabla u|$. By DCT(BCT),

$f(u_{k_j})\to f(u)$ in $L^1(\Omega')$, $f'(u_{k_j})\nabla u_{k_j}\to f'(u)\nabla u$ in $L^1(\Omega')$

$\Rightarrow f'(u_k)\nabla u_k\to f'(u)\nabla u$ in $L^1_{loc}(\Omega)$ By Thm1,

weak $\nabla f(u)=f'(u)\nabla u$

Thm2 If $u\in W^1(\Omega)$, then u^+, u^-, $|u|\in W^1(\Omega)$ and

$$\nabla u^+=\begin{cases}\nabla u, & u(x)>0\\ 0, & u(x)\le0\end{cases}\qquad \nabla u^-=\begin{cases}0, & u(x)\ge0\\ \nabla u, & u(x)<0\end{cases}\qquad \nabla|u|=\begin{cases}\nabla u(x), & u(x)>0\\ -\nabla u(x), & u(x)<0\\ 0, & u(x)=0\end{cases}$$

pf: $u^+ = \max\{u, 0\}$. For $\varepsilon > 0$, consider function

$$f_\varepsilon(t) = \begin{cases} (t^2+\varepsilon^2)^{\frac{1}{2}} - \varepsilon, & t > 0 \\ 0, & t \le 0 \end{cases} \Rightarrow f_\varepsilon(t) \in C^1(\mathbb{R}) \qquad \text{By Chain Rule,}$$

$$f_\varepsilon'(t) = \frac{t}{\sqrt{t^2+\varepsilon^2}} < 1$$

$\nabla f_\varepsilon(u) = f_\varepsilon'(u)\nabla u$. By Def of weak $\nabla f_\varepsilon(u)$: $\forall \varphi \in C_c^\infty(\Omega)$

$$\int_\Omega \nabla f_\varepsilon(u)\varphi\,dx = -\int_\Omega f_\varepsilon(u)\nabla\varphi\,dx = \int_\Omega \frac{u^+\nabla u}{\sqrt{u^2+\varepsilon^2}}\cdot\varphi\,dx = \int_{\{u>0\}} \frac{u^+\nabla u}{\sqrt{u^2+\varepsilon^2}}\cdot\varphi\,dx$$

$|f_\varepsilon(u)\nabla\varphi| \le u^+\cdot\nabla\varphi\cdot \mathbb{1}_{\operatorname{supp}\varphi} \in L^1(\Omega)$. By DCT,

$$\int_\Omega f_\varepsilon(u)\nabla\varphi\,dx \to \int_\Omega f(u)\nabla\varphi\,dx = \int_\Omega u^+\nabla\varphi\,dx \quad (\varepsilon\to 0)$$

$\left|\frac{u^+\nabla u}{\sqrt{u^2+\varepsilon^2}}\cdot\varphi\right| \le |\nabla u||\varphi|\cdot\mathbb{1}_{\operatorname{supp}\varphi} \in L^1(\Omega)$. By DCT.

$$\int_\Omega \frac{u\nabla u}{\sqrt{u^2+\varepsilon^2}}\cdot\varphi\mathbb{1}_{\{u>0\}}\,dx \to \int_\Omega \nabla u\cdot\varphi\mathbb{1}_{\{u>0\}}\,dx \quad (\varepsilon\to 0)$$

$$\Rightarrow \int_\Omega u^+\nabla\varphi\,dx = -\int_\Omega \nabla u\varphi\mathbb{1}_{\{u>0\}}\,dx \Rightarrow u^+ \in W^1(\Omega)$$

$u^- = -(-u)^+(x)$, $|u| = u^+ - u^-$

Cor1 $u \in W^1(\Omega)$, $\forall c \in \mathbb{R}$, let $\Omega_c = \{x \in \Omega : u(x) = c\}$. Then

weak $\nabla u = 0$ on Ω_c

pf: $\nabla u = \nabla(u-c) = \nabla[(u-c)^+ + (u-c)^-] = \nabla(u-c)^+ + \nabla(u-c)^- = 0$ on Ω_c

Def1 $1 \le p \le \infty$, $k \ge 0 \in \mathbb{N}$.

$$W^{k,p}(\Omega) = \{u \in W^k(\Omega) : \text{weak } \partial^\alpha u \in L^p(\Omega),\ \forall \alpha \in \mathbb{N}^n \text{ with } |\alpha| \le k\}$$

- $W^{k,p}(\Omega)$ is a linear space.

Def2 (Norm of $W^{k,p}(\Omega)$)

$1 \le p < \infty$: $\|u\|_{W^{k,p}} = \left[\int_\Omega \left(\sum_{|\alpha|=0}^k |\partial^\alpha u|^p\right)dx\right]^{\frac{1}{p}}$ $\Longleftrightarrow$ equivalent norm

$$\|\|u\|\|_{W^{k,p}} = \sum_{|\alpha|=0}^k \|\partial^\alpha u\|_{L^p(\Omega)}, \quad \|\partial^\alpha u\|_{L^p(\Omega)} = \left(\int_\Omega |\partial^\alpha u|^p dx\right)^{\frac{1}{p}}$$

Equivalent: By $\sum a_i^p \le \left(\sum a_i\right)^p$ $p \ge 1$, $a_i \ge 0$, $\|u\|_{W^{k,\infty}} = \sum_{|\alpha|=0}^k \|\partial^\alpha u\|_{L^\infty(\Omega)}$

$$\left[\int_\Omega \left(\sum_{|\alpha|=0}^k |\partial^\alpha u|^p\right)dx\right]^{\frac{1}{p}} \le \left[\sum_{|\alpha|=0}^k \left(\int_\Omega |\partial^\alpha u|dx\right)^p\right]^{\frac{1}{p}} \le$$